Sales and Inventory Analytics for Precast RCC Products

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project Title "Sales and Inventory Analytics for Precast RCC Products". I extend my appreciation to **"Laxmi Concrete Products"**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analysed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate

Name: Krishna Goel

Date: March 2,2025

1 Executive Summary and Title

The project is about the challenges faced by the business 'Laxmi Concrete Products,' a Precast RCC Product Trader. The business primarily operates in a B2B and B2C capacity, serving to Government & Private Contractors as well as to Individual Consumer.

The key challenges at ' Laxmi Concrete Product' arise due to ‘Moderately Unpredictable Demand’ leading to Inventory Pileup of Less Selling Items. Another key challenge arises due ‘Customers purchase on credit and delay the payments significantly or some even default’.

To find solution to above stated challenges, I aim to analyze the Sales Data to forecast the demand and also utilizing the Inventory Data along with it to manage the stock of less selling items. To tackle the problem of credit sales and customers defaulting on it, I aim to analyze the Customer Ledger Data. Through this analysis my goal is provide the business, a predictable sale range of their products for better inventory management and provide them a method to settle with credit sales.

2 Organization Background

Business Name: Laxmi Concrete Products

Proprietor Name: Mr. Sanjay Goel

Address: Opp. Green Field Public School, Village Sunped, Ballabgarh, Faridabad,

Haryana - 121004

Laxmi Concrete Products, a Precast RCC Product Trader established in 2016 by Mr. Sanjay Goel in Village Sunped, Ballabgarh. The Business Operates in both B2B and B2C capacity. The Business provides Precast RCC Products such as RCC Pipes, SFRC Manhole Covers and Frames, RCC Drain Covers, RCC Concrete Benches, Fencing Poles and other RCC Products to both Government and Private Projects.
Mr. Sanjay Goel is a Civil Engineer by profession and his vision is to provide Superior Quality Precast RCC products.
Laxmi Concrete Products is the Sister Concern of Laxmi Spun Pipes, which was initially founded by proprietor’s father in 1987. In 2016 Proprietor diversified the existing business by adding numerous new products and started this new Trading Firm. This well-rooted business has strong connections with other Businesses and Customers.

3 Problem Statement

1.1 MODERATELY UNPREDICTABLE DEMAND: Due to very nature of the business, owner faces a problem in forecasting actual demand leading to inefficient supply chain and inventory management.

1.2 PAYMENT DELAYS AND DEFAULTS: Business deals are frequently conducted on credit leading to delay in payments and often defaults leading to financial crunch eventually leading to lack of monetary liquidity needed at times.

4 Background of the Problem

The business has detailed sales and inventory data, but the problem of Moderately Unpredictable Demand is caused due to internal factors such as lack of analysis of that detailed data. Wide range of products with varying demand and their inventory requirement makes it difficult to predict the upcoming demand. The demand is also affected by some external factors like Market Trends which affect business due to imposition of new rules and regulations or emerging technologies.

In the industry of Precast RCC Products, where credit sales are very common due to long lasting industry relations, customer expects for extended credit terms which pressure business to relax their payment terms causing delay and defaults in payment. Also, proprietor does not have a robust system to decide whom to provide credit and my how much, leading to credit sales and payment relaxations beyond limit sometimes, even dealing with clients who default on their payments.

5 Problem Solving Approach

5.1: Details about the method used along with justification

For solving the problem of Moderately Unpredictable Demand I am using the approach of 'Time Series and Trend Analysis' which is used to predict future demand on the basis of the historical data. Analysing seasonal patterns and trends will result in understanding the variation in demand over time. This will help to maintain optimum amount of stock according to the demand and reduce the stocking of less selling items.

For solving the problem of Payment Delays and Defaults I am using the approach of the 'Cash Flow Analysis' on the Customer Ledger data. This will help in to identify the payment delay and defaults impact on overall cash flow of the business. Analysing this will help to implement strategy to improve the collection of cash, which will ensure timely collection, maintain cash flow and will reduce the overall financial risk to the business and help in establishing the criteria for credit sales eventually reducing the defaults and delays.

5.2: Details about the intended data collection with Justification

Sales Data: I collected the sales data from the firm that comprises of Product Details, total sale quantity for each product and total revenue generated by each product.

Analysing the past sales will help in identifying patterns, trends and demand resulting in properly forecasting the demand of the various products reducing the uncertainty of demand.

Inventory Data: This data comprises of product information, stock level of each product, stock movement and purchase order for products.

The Inventory Data will help to analyse the days of inventory outstanding, lead times, reorder point, safety stock. This will ensure that the stock management is aligned with the predicted demand and will also prevent stockouts and overstocks of any particular product.

Customer Ledger Data: Customer Ledger data provided me customer information, transaction details such as debit and credit entries and financial data of the business.

This will allow me to identify the payment delays and defaults that have happened and will allow me to set a benchmark for giving credit on sales. Also help to properly follow up with the customers for the collection of payments to maintain a proper cash flow.

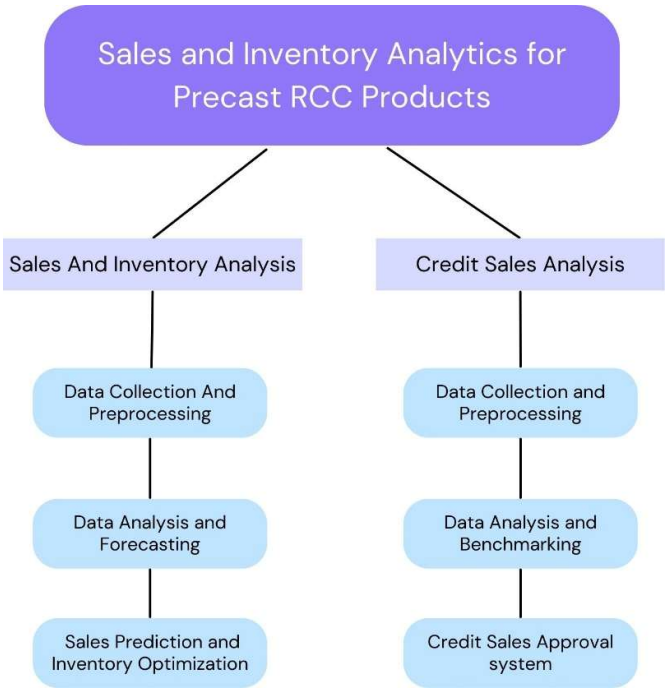
5.3: Details about the analysis tools with Justification

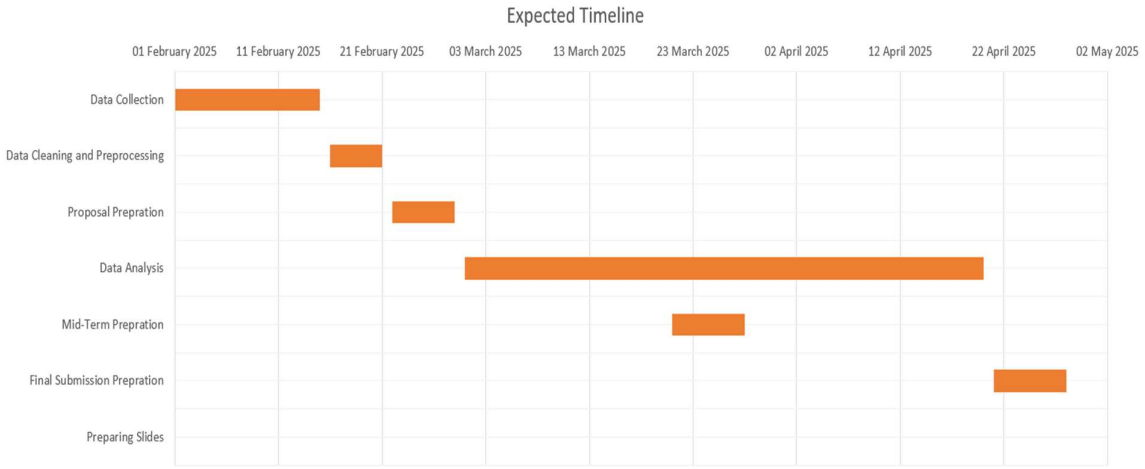
Analysis Tools: For the analysis of the Data collected by the business I aim to use Excel. I used Excel to clean and preprocess the data and will further analyse it using pivot tables, lookup and Charts and other functions provided by excel.

Justification: Excel provides pivot tables that will allow me to explore and identify key metrics of the data for better decision. Also charts and Graphs will help me to represent trends and patterns. The lookup function will help to retrieve data from large datasets.

6 Expected Timeline

6.1 Work Breakdown Structure:





7 Expected Outcome

- Demand Prediction will result in management of Inventory effectively that will prevent overstock and stockout for products.
- Demand Prediction will also result in increased customer satisfaction and trust as products will be readily available and can meet customer demands promptly.
- Establishing the benchmark for credit sales will help to identify whom to give credit and leading to proper cash flow and financial stability, and timely payments.
- Minimizing the risk of financial losses and instability by identifying high risk customers and adjusting the terms accordingly.
- This will help the business to take better decisions leading to efficiency and cost saving for them.